Limited-Display PCL Assortment Planning

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Abstract. Retail assortment planning converts a calibrated substitution model into a discrete display decision. Under the paired combinatorial logit model, every unordered product pair remains a choice nest when one or both products are offered. This candidate isolates that semantic requirement in a six-product, cardinality-constrained instance with a structured exact MILP and a complete enumeration oracle. The unique optimal assortment is $\{3, 5, 6\}$. Two independently generated programs implement different incorrect pair-state systems and both choose $\{2, 3, 5, 6\}$, whose correct expected margin is 11.64% below optimum. The programs solve successfully; the error lies entirely in translating the calibrated demand model.

Keywords. assortment optimization; paired combinatorial logit; discrete choice; revenue management; mixed-integer linear programming

Form domain: Revenue management and pricing • Model: MILP • Exact oracle: 57 feasible assortments

1. Business Brief

Business-level description. *Our category team must choose up to four products for a limited retail display. For each product, the demand-science team has estimated a customer-attraction weight and a normalized contribution margin. It has also calibrated a standard paired combinatorial logit (PCL) choice model, with a dissimilarity estimate for every unordered product pair and an outside-option weight for customers who leave without purchasing. Choose the assortment that maximizes expected contribution margin while accounting for substitution among the offered products.*

Naming the calibrated PCL engine makes the business request operationally precise while leaving the probability system and optimization formulation to the modeler.

2. PCL Choice System

Let N be the products, $x_i \in \{0, 1\}$ indicate whether product i is offered, $v_i > 0$ be attraction, $r_i \geq 0$ be normalized margin, $v_0 > 0$ be outside-option weight, and $\gamma_{ij} \in (0, 1]$ be pair dissimilarity. For pair $\{i, j\}$, define

$$I_{ij} = \left(v_i^{1/\gamma_{ij}} + v_j^{1/\gamma_{ij}}\right)^{\gamma_{ij}}, \tag{1}$$

$$\mu_{ij}^i = I_{ij}\frac{v_i^{1/\gamma_{ij}}}{v_i^{1/\gamma_{ij}} + v_j^{1/\gamma_{ij}}}, \qquad \mu_{ij}^j = I_{ij} - \mu_{ij}^i. \tag{2}$$

Use state binaries y_{ij}^{10}, y_{ij}^{01}, and y_{ij}^{11} for only i, only j, and both products offered. They satisfy

$$y_{ij}^{11} \leq x_i, \quad y_{ij}^{11} \leq x_j, \quad y_{ij}^{11} \geq x_i + x_j - 1, \tag{3}$$

$$y_{ij}^{10} = x_i - y_{ij}^{11}, \qquad y_{ij}^{01} = x_j - y_{ij}^{11}. \tag{4}$$

The standard PCL denominator and expected-margin numerator are

$$D(x) = v_0 + \sum_{i<j}\left(v_i y_{ij}^{10} + v_j y_{ij}^{01} + I_{ij} y_{ij}^{11}\right), \tag{5}$$

$$R(x) = \sum_{i<j}\left(r_i v_i y_{ij}^{10} + r_j v_j y_{ij}^{01} + (r_i \mu_{ij}^i + r_j \mu_{ij}^j) y_{ij}^{11}\right). \tag{6}$$

The exactly-one-offered states in Eqs. (5)–(6) are decisive: a pair vanishes only when neither member is offered.

3. Structured Exact MILP

The assortment problem maximizes $R(x)/D(x)$ subject to $\sum_i x_i \leq 4$. Apply a Charnes–Cooper transformation with $t = 1/D(x)$ and $w_{ij}^s = t y_{ij}^s$ for $s \in \{10, 01, 11\}$. Let b_{ij}^s and a_{ij}^s denote the denominator and numerator coefficients in Eqs. (5) and (6). Valid bounds $0 < \underline{t} \le t \le \bar{t}$ follow from $v_0 \le D(x) \le \overline{D}$. The exact MILP is

$$\max \quad \sum_{i<j} \sum_{s} a_{ij}^{s} w_{ij}^{s}, \tag{7}$$

$$\text{s.t.} \quad v_0 t + \sum_{i<j} \sum_{s} b_{ij}^{s} w_{ij}^{s} = 1, \tag{8}$$

$$\sum_{i \in N} x_i \le 4, \quad \text{Eqs. (3)–(4)}, \tag{9}$$

$$\underline{t} y_{ij}^{s} \le w_{ij}^{s} \le \bar{t} y_{ij}^{s} \qquad \forall i < j, s, \tag{10}$$

$$t - \bar{t}(1 - y_{ij}^{s}) \le w_{ij}^{s} \le t - \underline{t}(1 - y_{ij}^{s}) \qquad \forall i < j, s, \tag{11}$$

$$x_i, y_{ij}^{s} \in \{0, 1\}, \quad t, w_{ij}^{s} \ge 0. \tag{12}$$

Constraints (10)–(11) impose $w_{ij}^{s} = t y_{ij}^{s}$. The repository includes this structured Gurobi model and a separate exhaustive oracle over all 57 assortments of size at most four.

4. Exact Instance and Failure Certificate

The six products are derived from the official MIT-licensed correlated synthetic instance of Feng, Che, and Chen [1]. Product IDs are remapped, and $v_0 = 3.84538427435401$ preserves a no-purchase probability of 0.25 when all six products are displayed.

Table 1. Exact optimum and independently generated decisions.

Model	Selected products	Objective under that model
Correct standard PCL	$\{3, 5, 6\}$	0.323647238294198
Generated singleton-plus-pair model	$\{2, 3, 5, 6\}$	0.245658160940
Generated jointly-offered-pairs-only model	$\{2, 3, 5, 6\}$	0.188961826760

The first generated program adds one global singleton term per selected product, instead of one exactly-one-offered contribution for every absent counterpart. The second retains a pair nest only when both products are offered. Both programs are executable and prove their own MILP optima.

Failure certificate. The correct PCL oracle has the unique optimum $\{3, 5, 6\}$ with expected margin 0.323647238294198. Both generated programs choose $\{2, 3, 5, 6\}$. Evaluated under the calibrated standard PCL model, that decision is worth 0.285975423739680, giving absolute regret 0.037671814554518 and relative regret 11.639776%.

5. Reproducible Package

The package contains the exact derivative instance, source-product mapping, structured MILP, standard-library enumeration checker, ground truth, business brief, and upstream MIT notice. The instance embeds the upstream copyright and complete MIT permission notice, so every five-file Form package carries its rights chain. The full-data no-purchase probability, probability sums, objective, bound, and direct recomputation are checked mechanically. Public package: `https://github.com/Zhengzhong-You/OR-Bench`.

References

[1] J. Feng, A. Che, and Z.-L. Chen. 2026. Assortment optimization with general linear constraints under the paired combinatorial logit choice model. *INFORMS Journal on Computing*, Articles in Advance. doi:10.1287/ijoc.2024.0830.

[2] INFORMS Journal on Computing Software and Data Repository. 2026. Artifact for Feng, Che, and Chen. doi:10.1287/ijoc.2024.0830.cd.